Review plan · for the teacher

# Protecting models and data: unit check

AI security · Level 3 · Unit check

| Time | Materials |
| --- | --- |
| 45 min | Board, chalk, paper. No devices needed. |

## Learning goals

- Poisoned data
- Models that remember too much
- Pictures that fool a model
- Is this the real model?

## Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

## Discussion

- Which check will you start with at your school?
- Which question surprised you, and why?
- How would you explain a fingerprint to a colleague?

## Exit ticket

- One check I will start using, and where

**Try it at school tomorrow**
Pick the check from this unit your school needs most (median checks on class data, redacting before sharing, a person deciding what matters, or a list of AI tools with their sources), start it this week, and tell a colleague how it went.

Quiz sheet · for each team

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Team _____ Date _____

**1** **What is data poisoning?**

◯ Breaking into the computer that runs the model ◯ Slipping wrong or planted examples into training data
◯ Asking a model too many questions at once

**2** **A backdoor makes a model misbehave only when a hidden trigger appears, so normal testing rarely finds it.**

◯ True ◯ False

**3** **What does this program print?**

```
data = [50, 52, 49, 51, 5]
m = sorted(data)[len(data) // 2]
for x in data:
    if abs(x - m) > m / 2:
        print("check", x)
```

◯ check 5 ◯ check 52 ◯ Nothing

**4** **A school wants to train a homework helper on pupils' past work. Which steps protect pupils' privacy? Choose all that apply.**

☐ Leave out work the helper doesn't need ☐ Test with a planted canary before anyone uses it
☐ Remove names, numbers and rare details first ☐ Train on everything, then delete the original files

**5** **Match each risk to a defence.**

| | | |
|---|---|---|
| Poisoned training data | • • • • | Test with small changes; a person decides |
| A model that memorises private details | • • • • | Check the fingerprint and signature |
| Adversarial examples | • • • • | Know the source; keep a clean test set |
| A tampered or fake model file | • • • • | Keep personal data out of training |

**6** **What is membership inference?**

◯ Guessing whether someone's data was used to train a model ◯ Adding new members to a school's AI club
◯ Checking a model file's fingerprint

**7** **A model that says it is 99% sure can still be fooled by a tiny change to its input.**

◯ True ◯ False

**8** **Put in order: using a model file you downloaded.**

☐ Compare with the fingerprint the maker published
☐ Add it to the school's list of tools
☐ Get it from the maker's official source
☐ Compute its SHA-256 fingerprint

**9** **What does a valid signature tell you that a matching fingerprint alone doesn't?**

◯ How big the file is ◯ How accurate the model is ◯ Who published the file

**10** **Complete the sentence.**

A fingerprint shows the file is _____; a signature shows _____ made it.

**Word bank:** smaller · unchanged · when · who

Answers · for the teacher

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1 **Slipping wrong or planted examples into training data**

Poisoning spoils what the model learns from.

2 **True**

Ordinary inputs never contain the trigger, so the model looks normal in ordinary tests.

3 **check 5**

Sorted, the values are 5, 49, 50, 51, 52, so the median is 50.

4 **Leave out work the helper doesn't need; Remove names, numbers and rare details first; Test with a planted canary before anyone uses it**

Keeping data out and redacting what must go in reduce leaks, and a canary test shows whether any remain.

5 **Poisoned training data — Know the source; keep a clean test set**
**A model that memorises private details — Keep personal data out of training**
**Adversarial examples — Test with small changes; a person decides**
**A tampered or fake model file — Check the fingerprint and signature**

Each risk in this unit has its own defence: guard the data, keep personal data out, test and keep a person deciding, and check where files came from.

6 **Guessing whether someone's data was used to train a model**

An attacker checks how familiar a record feels to a model and guesses whether it was in the training data.

7 **True**

Confidence is a number the model gives itself, not a guarantee.

8 **Top to bottom: 3, 4, 1, 2**

Source first, because the published fingerprint is only as trustworthy as where it came from; then compute, compare, and record what the school runs.

9 **Who published the file**

A fingerprint shows the bytes didn't change; an impostor can publish a matching one for a fake file.

10 **A fingerprint shows the file is unchanged; a signature shows who made it.**

Fingerprint: the bytes arrived unchanged.